PREVIOUS PAPER Held on: 19-11-2016

RRB Assistant (PT)

(Based on memory)

Test-I: Reasoning Ability

Directions (1-5): Study the following information carefully to answer the given questions.

Eight friends A, B, C, D, E, F, G and H are sitting around a circle facing the centre. A sits third to the left of B, and second to the right of F. D does not sit next to A or B. C and G always sit next to each other. H never sits next to D and C does not sit next to B.

1. Which of the following pairs sits between H and E?
1) F, D 2) H, B 3) G, G 4) E, G 5) None of these
2. Starting from A's position, if all the eight friends are arranged in alphabetical order in clockwise direction the seating positions of how many members, excluding A, do not change?
1) None 2) One
3) Two 4) Three
5) Other than those given as options
3. Which of the following pairs has only one person sitting between them, if the counting is done in clockwise direction?
1) A, B 2) C, D
3) F, E 4) G, H
5) Other than those given as options
4. Who sits on the immediate right of E?
1) A 2) D
3) F 4) H
5) Other than those given as options
5. What is the position of B with respect to C?
1) Second to the left 2) Third to the right
3) Third to the left 4) Can't be determined
5) Other than those given as options

Directions (Q. 6-10): In this question, relationship between different elements is shown in the statements. The statements are followed by conclusions. Study the conclusions based on the given statement and select the appropriate answer.

6. **Statement:** $H = W \leq R > F$
Conclusions: **I.** $R = H$ **II.** $R > H$
1) if only conclusion I is true
2) if only conclusion II is true
3) if either conclusion I or II is true
4) if neither conclusion I nor II is true
5) if both conclusion I and II are true
7. **Statement:** $M < T > K = D$
Conclusions: **I.** $D < T$ **II.** $K < M$
1) If both conclusion I and II are true
2) If either conclusion I or II is true
3) If only conclusion I is true
4) If neither conclusion I nor II is true
5) If only conclusion II is true
8. **Statement:** $R \leq N \geq F > B$
Conclusions **I.** $F = R$ **II.** $B < N$
1) Neither conclusion I nor II is true
2) Both conclusion I and II are true
3) Only conclusion II is true
4) Only conclusion I is true
5) Either conclusion I or II is true
9. **Statement:** $H > W < M \geq K$
Conclusions: **I.** $K < W$ **II.** $H > M$
1) Only conclusion II is true
2) Only conclusion I is true
3) Neither conclusion I nor II is true
4) Both conclusion I and II are true
5) Either conclusion I or II is true
10. **Statement:** $R \geq T = M > D$
Conclusions: **I.** $D < T$ **II.** $R \geq M$
1) Either conclusion I or II is true
2) Both conclusion I and II are true
3) Neither conclusion I nor II is true
4) Only conclusion I is true
5) Only conclusion II is true

Directions (Q. 11-15): Study the following information carefully and answer the questions given below:

E 4 B % R 3 A 6 # F H @ I 2 D 9 © K U S W 1 M P 5 • Q 8 T

11. If all the numbers are dropped from the above arrangement, which of the following will be ninth to the left of W?
1) A 2) #
3) R 4) ©
5) Other than those given as options
12. How many such numbers are there in the given arrangement each of which is immediately preceded by a symbol and immediately followed by a letter?
1) None 2) Two 3) Three
4) More than three 5) One
13. Which of the following is fifth to the right of the eighteenth from the right end of the above arrangement?
1) © 2) I
3) A 4) M
5) Other than those given as options

14. Which of the following is fourth to the right of twelfth from the left end of the above arrangement?
1) 2
2) D
3) 9
4) M
5) Other than those given as options

15. Four of the following are alike in a certain way based on their positions in the given arrangement and hence form a group. Which is the one that does not belong to that group?
1) F@#
2) D©2
3) UWK
4) %3B
5) 5QM

Directions (Q. 16-20): In each of these questions two/three statements followed by two conclusions numbered I and II have been given. You have to take the given statements to be true even if they seem to be at variance with commonly known facts and then decide which of the given conclusions logically follows. Give answer
1) if conclusion I follows.
2) if conclusion II follows.
3) if either conclusion I or II follows.
4) if neither conclusion I nor II follows.
5) if both conclusion I and II follow.

16. **Statements:** No tea is coffee.
No sweet is tea.
Conclusions: **I.** No coffee is sweet.
II. All sweets are coffee.

17. **Statements:** All medals are awards.
All rewards are medals.
Conclusions: **I.** All rewards are awards.
II. All awards are medals.

18. **Statements:** Some leaves are plants.
All bushes are plants.
Conclusions: **I.** At least some leaves are bushes.
II. Some leaves are definitely not bushes.

19. **Statements:** All bottles are mugs.
No cup is a mug.
Conclusions: **I.** No bottle is a cup.
II. At least some mugs are bottles.

20. **Statements:** All windows are doors.
All entrances are windows.
No gate is a door.
Conclusions: **I.** At least some windows are gates.
II. No gate is an entrance.

Directions (Q. 21-25): Study the following information carefully and answer the questions given below:

A, B, C, D, E, F and G are sitting in a straight line facing north, but not necessarily in the same order. There is only one person between F and C. E sits between A and D. There are only two persons between E and G. F sits on the immediate left of A, who sits in the middle of the row.

21. How many persons are there between E and F?
1) One
2) Two
3) Three
4) Can't be determined
5) Other than those given as options

22. Who among the following sit at the extreme ends of the row?
1) D, F
2) G, C
3) B, C
4) Can't be determined
5) Other than those given as options

23. Who among the following sits on the immediate right of D?
1) G
2) E
3) F
4) B
5) None of these

24. Who among the following sits third to the right of A?
1) C
2) G
3) B
4) E
5) Other than those given as options

25. Which of the following statements is true with regard to B?
1) B is second to the right of A.
2) B is fourth to the left of G.
3) B sits at the extreme right end of the row.
4) B sits at the extreme left end of the row.
5) Other than those given as options

26. The positions of how many digits in the number 59164823 will remain unchanged after the digits are rearranged in descending order within the number?
1) None
2) One
3) Two
4) Three
5) More than three

27. What should come next in the following letter series based on English alphabet?
CEA IKG OQM ?
1) STW
2) WUS
3) SWU
4) UWS
5) Other than those given as options

28. In a row of 40 children facing North, E is eighth to the right of V. If V is 18th from the right end of the row, what is the position of E from the left end of the row?
1) 32nd
2) 10th
3) 31st
4) 29th
5) Other than those given as options

Directions (Q. 29-33): Following questions are based on the five three-digit numbers given below:

853 581 747 474 398

29. If all the digits in each of the numbers are arranged in descending order within the number, which of the following will form the lowest in the new arrangement of the numbers?
1) 853
2) 581
3) 747
4) 398
5) 474

30. If all the numbers are arranged in ascending order from left to right, which of the following will be the sum of all three digits of the number which is exactly in the middle of the new arrangement?
1) 17
2) 15
3) 14
4) 13
5) 19

31. What will be the resultant if the third digit of the lowest

number is multiplied by the second digit of the highest number?

1) 27 2) 40 3) 20 4) 45 5) 19

32. If the positions of the second and the third digit of each of the numbers are interchanged, how many even numbers will be formed?

1) None 2) One 3) Two 4) Three 5) Four

33. If one is added to the first digit of each of the numbers, how many numbers thus formed will be divisible by three?

1) None 2) One 3) Two 4) Three 5) Four

34. In a certain code language JANUARY is written as ZSBTOBK. How is OCTOBER written in that code language?

1) SFCPUDP 2) SFCNUDP
3) SCFNDUP 4) FSCNUDP
5) Other than those given as options

Directions (Q. 35-37): Study the following information carefully to answer the given questions:

B is sister of A. A is father of G. H is the only son of F. F is only son-in-law of A. G is mother of H.

35. If C is husband of B, then how is A related to C?

1) Father 2) Brother-in-law 3) Mother
4) Brother 5) None of these

36. How is G related to B?

1) Brother 2) Niece
3) Sister 4) Nephew
5) Other than those given as options

37. How is A related to H?

1) Uncle
2) Father
3) Paternal grandfather
4) Maternal grandfather
5) Other than those given as options

Directions (Q. 38-39): Study the following information carefully to answer the questions.

A vehicle starts from point P and runs 10 km towards North. It takes a right turn and runs 15 km. Now it runs 6 km after taking a left turn. Finally, it takes a left turn, runs 15 km and stops at point Q.

38. How far is point Q with respect to point P?

1) 16 km 2) 25 km
3) 4 km 4) 10 km
5) Other than those given as options

39. Towards which direction was the vehicle moving before stopping at point Q?

1) North 2) East 3) South
4) West 5) Northwest

40. In a row of 34 students, W is fifth after X from the front and X is 20th from the back. What is the position of W from the front?

1) 20 2) 25
3) 30 4) 22
5) Other than those given as options

Test-II: Quantitative Aptitude

Directions (Q. 41-45): What will come in place of question mark (?) in the following number series?

41. 12 13 17 26 42 ?

1) 57 2) 58
3) 59 4) 75
5) Other than those given as options

42. 1 2 8 48 384 ?

1) 3440 2) 3840
3) 3820 4) 3550
5) Other than those given as options

43. 157 150 136 115 87 ?

1) 50 2) 51
3) 52 4) 54
5) Other than those given as options

44. 1 4 18 44 83 ?

1) 131 2) 132
3) 135 4) 136
5) Other than those given as options

45. 8 4 4 6 12 ?

1) 30 2) 34
3) 38 4) 42
5) Other than those given as options

Directions (Q. 46-60): What will come in place of question mark (?) in the following questions?

46. $80.137 \times 9 + 2.11 \times 139.7 = ?$

1) 916 2) 1016 3) 1216 4) 1026 5) 1256

47. $7802 + 132 - 8963 + 1326 = ? \times 33$

1) 6 2) 12 3) 21 4) 9 5) 14

48. 21.9% of 650 = ? + 23.12

1) 121.23 2) 109.23
3) 119.32 4) 129.23
5) Other than those given as options

49. $6666 \div 66 \div 0.25 = ?$

1) 101 2) 404
3) 304 4) 40.4
5) Other than those given as options

50. $\sqrt{?} + 18 = \sqrt{2704}$

1) 1256 2) 1156 3) 1296 4) 1024 5) 1466

51. $217 + 435 - 317 + 5110 = ?$

1) 9710 2) 7710
3) 8710 4) 8470
5) Other than those given as options

52. $164 \times 43 - 6070 = ?$

1) 682 2) 792 3) 882 4) 1082 5) 982

53. 14.5% of 740 − ?% of 320 = 87.3

1) 6.75 2) 6.25 3) 12.5 4) 14.75 5) 8.25

54. $(27)^3 \times 3^4 \div (81)^2 = 3^?$

1) 2 2) 5
3) 4 4) 3
5) Other than those given as options

55. $37.135 \text{ of } 25 + 125 \text{ of } 1.061 = \sqrt{?} + 894$
1) 28899 2) 29899
3) 27789 4) 27889
5) Other than those given as options

56. $4376 + 3209 - 1784 + 97 = 3125 + ?$
1) 2713 2) 2743 3) 2773 4) 2793 5) 2737

57. $\sqrt{?} + 14 = \sqrt{2601}$
1) 1521 2) 1369 3) 1225 4) 961 5) 1296

58. 85% of 420 + ?% of 1080 = 735
1) 25 2) 30 3) 35 4) 40 5) 45

59. $3024 \div 54 \text{ of } 19 - 84 = ?$
1) 920 2) 940 3) 960 4) 980 5) 840

60. 30% of 1225 – 64% of 555 = ?
1) 10.7 2) 12.3
3) 13.4 4) 17.5
5) Other than those given as options

Directions (Q. 61-65): Study the following table and answer the questions given below.

Number of tourists who visit different cities by different modes of transport

Cities	Vehicles				
	Car	Train	Bus	Bike	By Air
Delhi	192	188	172	191	174
Mumbai	180	166	178	187	182
Chandigarh	156	194	163	181	148
Dehradun	132	185	142	170	148
Mussoorie	149	159	155	149	183
Jaipur	168	163	158	142	174

61. What the average number of tourists who come by Train?
1) 190.5 2) 188.5
3) 175.83 4) 137.5
5) Other than those given as options

62. What is the difference between the total number of tourists who went to Mumbai and that to Mussoorie by all the vehicles?
1) 78 2) 98
3) 88 4) 83
5) Other than those given as options

63. What is the percentage of tourists who went to Dehradun by Train compared to the number of tourists who went to Chandigarh by Air?
1) 125 2) 145
3) 137 4) 160
5) Other than those given as options

64. What is the difference between the average number of tourists who went by Air to the average number of tourists who went by Bus?
1) 7.58 2) 9.97
3) 6.83 4) 2.30
5) Other than those given as options

65. What is the ratio of the number of tourists to Delhi who went by Car to that to Mumbai who went by Air?
1) 35 : 83 2) 45 : 71
3) 96 : 91 4) 32 : 7
5) Other than those given as options

66. If the wheel of a bicycle makes 560 revolutions in travelling 1.1 km, what is its radius? (use $\pi = \frac{22}{7}$)
1) 31.25 cm 2) 37.75 cm
3) 35.15 cm 4) 11.25 cm
5) Other than those given as options

67. Elena's age after 15 years will be 5 times her age 5 years back. What is her present age?
1) 10 years 2) 37 years
3) 35 years 4) 11 years
5) Other than those given as options

68. A man purchased a cow for ₹3000 and sold it the same day for ₹3600, allowing the buyer a credit of 2 years. If the rate of interest be 10% per annum, then the man has a gain of
1) 5% 2) 0%
3) 20% 4) 10%
5) Other than those given as options

69. A man takes 3 hours 45 minutes to row a boat 15 km downstream a river and 2 hours 30 minutes to cover a distance of 5 km upstream. Find the speed of the stream.
1) 1 kmph 2) 3 kmph
3) 5 kmph 4) 2 kmph
5) Other than those given as options

70. A cistern 6 m-long and 4-m wide contains water up to a height of 1 m 25 cm. Find the total area of the wet surface of the cistern.
1) 42 sqm 2) 49 sqm
3) 52 sqm 4) 64 sqm
5) Other than those given as options

71. In terms of percentage profit, which of following is the best transaction?
1) CP 36, Profit 17 2) CP 50, Profit 24
3) CP 40, Profit 19 4) CP 60, Profit 29
5) CP 45, Profit 21

72. The milk and water in two vessels A and B are in the ratio of 4 : 3 and 2 : 3 respectively. In what ratio should the liquids in both the vessels be mixed to obtain a new mixture in vessel C consisting of half milk and half water?
1) 8 : 3 2) 7 : 5
3) 4 : 3 4) 2 : 3
5) Other than those given as options

73. The average price of 10 books is ₹12 while the average price of 8 of these books is ₹11.75. Of the remaining two books, if the price of one book is 60% more than

the price of the other, what is the price of each of these two books?
1) ₹5, ₹7.50
2) ₹8, ₹12
3) ₹10, ₹16
4) ₹12, ₹14
5) Other than those given as options

74. A fort has provisions for 60 days. If after 15 days 500 men strengthen them and the food lasts 40 days longer, then how many men are there in the fort?
1) 3500
2) 4000
3) 6000
4) 8000
5) Other than those given as options

75. If a commission of 10% is given on the marked price of a truck the dealer gains 20%. If the commission is increased to 15%, the gain % will be how much?
1) $\frac{40}{3}$
2) 10
3) 20
4) 15
5) Other than those given as options

76. If a carton containing a dozen of mirrors is dropped, then which of the following cannot be the ratio of the broken mirrors to the unbroken mirrors?
1) 7 : 5
2) 3 : 1
3) 3 : 2
4) 2 : 1
5) Cannot be determined

77. A bag contains ₹216 in the form of ₹1, 50-paisa and 25-paisa coins in the ratio of 2 : 3 : 4. The number of 50-paisa coins is
1) 140
2) 175
3) 184
4) 160
5) 144

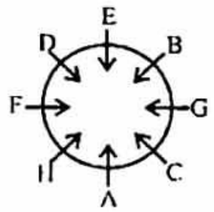
78. A is twice as fast as B and B is thrice as fast as C. The distance covered by C in 42 min will be covered by B in
1) 14 min
2) 4 min
3) 5 min
4) 8 min
5) 6 min

79. The CP of two dozen mangoes is ₹32. After selling 18 mangoes at ₹12 per dozen, the shopkeeper reduced the rate to ₹4 per dozen. Then find the loss percentage.
1) 15
2) 20
3) 25
4) 37.5
5) Other than those given as options

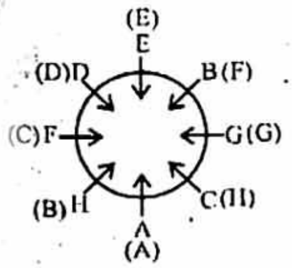
80. How many kilograms of sugar costing ₹9 per kg must be mixed with 27 kg of sugar costing ₹7 per kg so that there may be a gain of 10% by selling the mixture at ₹9.24 per kg?
1) 60 kg
2) 63 kg
3) 50 kg
4) 77 kg
5) Other than those given as options

Answers

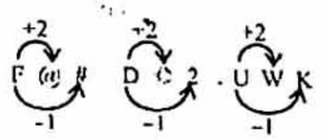
(1-5):

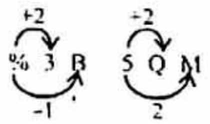


1. 1
2. 4;



Thus, there are three such members – D, E and G.
3. 3; FE is only such pair.
4. 2
5. 5; Second to the right
6. 3; **Given statement:**
H = W ≤ R > F
Check for conclusion I and II. Thus, H ≤ R is true. It means H = R or H < R may be true. Thus both conclusion I (R = H) and conclusion II (R > H or H < R) make a complementary pair. Hence either conclusion I or II is true.
7. 3; **Given statement:** M < T > K = D
Check for conclusion I. T > D or D < T is true.
Check for conclusion II. We can't compare K and M. Hence II (K < M) is not true.
8. 3; **Given statement:** R ≤ N ≥ F > B
Thus, we can't compare F and R. Hence I (F = R) is not true. Again, N > B or B < N is true.
Hence only conclusion II is true.
9. 3; **Given statement:**
H > W < M ≥ K
Thus, we can't compare W and K. Hence I (K < W) is not true.
Again, we can't compare H and M. Hence II (H > M) is also not true. So, neither conclusion I nor II is true.
10. 2; **Given statement:**
R ≥ T = M > D
Check for conclusion I. Thus, T > D or D < T is true.
Check for conclusion II. R ≥ M is true.
11. 5; The new arrangement becomes:
E B % R A # F H @ I D © K U $ W M P • Q T
Now, ninth to the left of W is F.
12. 1; Symbol | Number | Letter
Thus, there is no such number
13. 1; Fifth to the right of eighteenth from the right = (18 – 5 =)13th from the right end = ©.
14. 3; Fourth to the right of twelfth from the left end = (12 + 4 =) 16th from the left end = 9.
15. 5;





16. 4; No sweet is tea (E) + No tea is coffee (E) = E + E = No conclusion. Hence neither conclusion I nor II follows.
17. 1; All rewards are medals (A) + All medals are awards (A) = A + A = A = All rewards are awards (A). Hence conclusion I follows.
Again, All medals are awards (A) → conversion → Some awards are medals (I). Hence conclusion II does not follow.
18. 3; All bushes are plants (A) + (Some leaves are plants (I) → conversion →) Some plants are leaves (I) = A + I = No conclusion. But conclusion I and II make a complementary pair (I-O). Hence either I or II follows.
19. 5; All bottles are mugs (A) + (No cup is a mug (E) → conversion →) No mug is a cup (E) = A + E = E = No bottle is a cup. Hence conclusion I follows.
Again, All bottles are mugs (A) → conversion → Some mugs are bottles (I). Hence II also follows.
20. 2; All entrances are windows (A) + All windows are doors (A) = A + A = All entrances are doors (E) + (No gate is a door) → conversion →) No door is a gate (E) = A + E = E = No entrance is a gate (E) → conversion → No gate is an entrance (E). Hence conclusion II follows.
Again, All windows are doors (A) + No door is a gate (E) = A + E = E = No window is a gate. Hence conclusion I does not follow.

RRB Assistant PT (Previous Paper)

(21-25):

↑Facing north ↑C ↑G ↑F ↑A ↑E ↑D ↑B

21. 1 **22.** 3 **23.** 4 **24.** 3 **25.** 3

26. 2; Given number 5 9 1 6 $\boxed{4}$ 8 2 3

9 8 6 5 $\boxed{4}$ 3 2 1

Thus, there is only one such digit.

27. 4;

+2 +8 +2 +8 +2 +8 +2

C E A I K G O O M U W S

−4 −4 −4 −4

28. 3; V is 18th from the right end.
E's position from the right end
= (18 − 8 =) 10th
E's position from the left end
= (40 − 10 + 1 =) 31st

29. 5; Original numbers: 853 581 747 474 398
New numbers become: 853 851 774 744 983. Thus, the lowest number will be 744, which comes from 474.

30. 3; New arrangement becomes:
398 474 581 747 853
The middle number of the arrangement is 581.
∴ Sum of the digits of the number 581
= 5 + 8 + 1 = 14

31. 2; The lowest number is 398.
∴ Third digit of the lowest number is 8.
The highest number is 853.
∴ Second digit of the highest number is 5.
∴ Resultant = 8 × 5 = 40

32. 3;

Original number: 853 581 747 474 398

After changing new numbers become: 835 518 774 447 389

Thus, there are two even numbers formed, ie 518 and 774.

33. 3;

Original number: 853 581 747 474 398

+1 −1 +1 +1 −1

Now, new numbers become 953 681 847 574 498

In any number if the sum is a multiple of 3 it will be divisible by 3.
There are only two such numbers whose sum is a multiple of 3.
ie 681 = 6 + 8 + 1 = 15
And 498 = 4 + 9 + 8 = 21

34. 2; JANUARY → Reverse the word.

Y R A U N A J

+1 +1 +1 +1 +1 −1 +1

Z S B V O B K

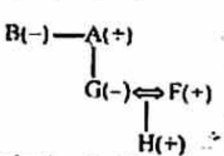
Similarly, OCTOBER → Reverse the word.

R E B O T C O

+1 +1 +1 −1 +1 +1 +1

S F C N U D P

(35-37):

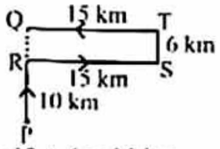


35. 2; If C is husband of B, then A is brother-in-law of C.

36. 2

37. 4

(38-39):



38. 1; QP = 10 + 6 = 16 km

39. 4

40. 1; X's position from front
= (34 − 20 + 1) = 15th
W's position from front
= (15 + 5 =) 20th

41. 5; The series is $+1^2, +2^2, +3^2, +4^2, +5^2, ...$
ie $12 + 1^2 = 13$, $13 + 2^2 = 17$, $17 + 3^2 = 26$, $26 + 4^2 = 42$, $42 + 5^2 = 67$, ...

42. 2; The series is ×2, ×4, ×6, ×8, ×10, ...
ie 1 × 2 = 2, 2 × 4 = 8, 8 × 6 = 48,
48 × 8 = 384, 384 × 10 = 3840, ...

43. 3; The series is −7, −14, −21, −28, ...
ie 157 − 7 = 150, 150 − 14 = 136,
136 − 21 = 115, 115 − 28 = 87, 87 − 35 = 52, ...

44. 4; The series is

1 4 18 44 83 136

+3 +14 +26 +39 +53

+11 +12 +13 +14

45. 1; The series is ×0.5, ×1, ×1.5, ×2, ...
ie 8 × 0.5 = 4, 4 × 1 = 4, 4 × 1.5 = 6,
6 × 2 = 12, 12 × 2.5 = 30, ...

46. 2; ? = 80.137 × 9 + 2.11 × 139.7
= 721.233 + 294.767 = 1016

47. 4; 7802 + 132 − 8963 + 1326 = ? × 33
or, 297 = ? × 33

$\therefore ? = \frac{297}{33} = 9$

48. 5; ? + 23.12 = 21.9% of 650
Solving by breaking method,
? = 21% of 650 + 0.9% of 650 − 23.12
= 21 × 6.50 + 0.9 × 6.50 − 23.12 = 136.5 + 5.85 − 23.12 = 142.35 − 23.12 = 119.23

49. 2; ? = 6666 ÷ 66 ÷ 0.25

$= \frac{6666}{66 \times 0.25} = \frac{101}{0.25} = \frac{10100}{25} = 404$

50. 2; $\sqrt{?} + 18 = \sqrt{2704} = 52$

or, $\sqrt{?} = 52 - 18 = 34$

∴ ? = 34 × 34 = 1156

51. 5; ? = 217 + 435 − 317 + 5110
Units digit = 7 + 5 + 0 − 7 = 5
Tens digit = 1 + 3 + 1 − 1 = 4
Hundreds digit = 2 + 4 + 1 − 3 = 4
Thousands digit = 5
Reqd answer = 5445

52. 5; ? = 164 × 43 − 6070
= 164 × (40 + 3) − 6070
= 6560 + 492 − 6070 = 982

53. 2; 14.5% of 740 − ?% of 320 = 87.3

or, $\frac{? \times 320}{100}$ = 14.5% of 740 − 87.3

= 14.5 × 7.40 − 87.3
= 107.3 − 87.3 = 20

$\therefore ? = \frac{20 \times 100}{320} = \frac{100}{16} = 6.25$

54. 2; $(27)^3 \times 3^4 \div (81)^2 = 3^?$

or, $\frac{3^{3\times3} \times 3^4}{3^{4\times2}} = 3^?$ or, $3^{9+4-8} = 3^?$

or, $3^? = 3^{1+4} = 3^5$ ∴ ? = 5

55. 4; 37.135 of 25 + 125 of 1.061 = $\sqrt{?}$ + 894

or, $\sqrt{?}$ = 37.135 × 25 + 125 × 1.061 − 894
= 928.375 + 132.625 − 894
= 1061 − 894 = 167
∴ ? = 167 × 167 = 27889

56. 3; 4376 + 3209 − 1784 + 97 = 3125 + ?
or, ? = 4376 + 3209 − 1784 + 97 − 3125
= 2773

57. 2; $\sqrt{?} + 14 = \sqrt{2601}$

or, $\sqrt{?} = 51 - 14 = 37$

∴ ? = 37 × 37 = 1369

58. 3; 85% of 420 + ?% of 1080 = 735
or, ?% of 1080 = 735 − 85% of 420

or, $\frac{? \times 1080}{100} = 735 - \frac{85 \times 420}{100}$

$= 735 - \frac{17 \times 420}{20} = 735 - 17 \times 21$

= 735 − 357 = 378

$\therefore ? = \frac{378 \times 100}{1080} = \frac{3780}{108} = 35$

59. 4; ? = 3024 ÷ 54 of 19 − 84
= 56 × 19 − 84 = 1064 − 84 = 980

60. 2; ? = 30% of 1225 − 64% of 555
= 30 × 12.25 − 64 × 5.55
= 367.5 − 355.2 = 12.3

61. 3; Reqd average

$= \frac{188+166+194+185+159+163}{6} = \frac{1055}{6}$

= 175.83

62. 2; Reqd difference = (180 − 149) + (166 − 159) + (178 − 155) + (187 − 149) + (182 − 183) = 31 + 7 + 23 + 38 − 1 = 99 − 1 = 98

63. 1; Reqd % = $\frac{185}{148} \times 100 = 125\%$

64. 3; Reqd average difference

$= \frac{1}{6}$ {(174 − 172) + (182 − 178) + (148 − 163) + (148 − 142) + (183 − 155) + (174 − 158)}

$= \frac{1}{6}$ {2 + 4 − 15 + 6 + 28 + 16}

$= \frac{1}{6} \{41\} = \frac{1}{6} \times 41 = 6.83$

65. 3; Reqd ratio = $\frac{192}{182} = \frac{96}{91} = 96:91$

66. 1; No. of revolutions

$= \frac{\text{Distance}}{2\pi r}$

$\therefore r = \frac{1.1 km \times 7}{2 \times 22 \times 560} = \frac{110000 \times 7}{2 \times 22 \times 560}$
$= 31.25$ cm

67. 1; Let Elena's age be x years.
Now, $x + 15 = 5(x - 5)$
or, $x + 15 = 5x - 25$
or, $4x = 40$
$\therefore x = \frac{40}{4} = 10$ years

68. 2; Interest $= 3600 - 3000 = 600$
Let the amount after two years be P.
Then, $\frac{P \times 10 \times 2}{100} = 600$
$\therefore P = 600 \times 5 = 3000$
Gain $= 3000 - 3000 = 0\%$

Method II. CP = ₹3000, SP $= \frac{3600 \times 100}{100 + 10 \times 2}$
= ₹3000
$\therefore$ Gain% $= 3000 - 3000 = 0\%$

69. 1; Let the speed of the [illegible] and that of boat be x kmph.
Then, $\frac{15}{x+y} = 3\frac{45}{60} = 3\frac{3}{4} = \frac{15}{4}$
$\therefore x + y = 4$ ... (i)
Again, $\frac{5}{x-y} = 2\frac{1}{2} = \frac{5}{2}$
or, $x - y = 2$ ... (ii)
Solving (i) and (ii), we get
$2x = 6$
$\therefore x = 3$ km
$\therefore y = 1$ kmph

70. 2; Area of the wet surface
$= 2(lb + bh + lh) - lb$
$= lb + 2bh + 2lh$
$= 6 \times 4 + 2 \times 4 \times 1.25 + 2 \times 6 \times 1.25$
$= 24 + 10 + 15 = 49$ sq m

71. 4; We check the fraction value of each options.
1) $\frac{17}{36} = 0.47$
2) $\frac{24}{50} = 0.48$
3) $\frac{19}{40} = 0.475$
4) $\frac{29}{60} = 0.4833$
5) $\frac{21}{45} = 0.46$
Hence the best transaction in terms of percentage is option 4).

72. 2; In vessel A, milk $= \frac{3}{7}$ of the weight of the mixture.
In vessel B, milk $= \frac{2}{5}$ of the weight of the mixture.
Now, we want to form a mixture which will be $\frac{1}{2}$ of the weight of the mixture.

By alligation method

$\frac{3}{7}$ $\quad$ $\frac{2}{5}$
$\frac{1}{2}$
$\frac{1}{10}$ $\quad$ $\frac{1}{14}$

$\therefore$ Reqd ratio $= 14 : 10 = 7 : 5$

73. 3; Average price of the remaining two books $= 12 \times 10 - 8 \times 11.75 = 120 - 94 = 26$
Now, let the price of one book be x.
Then that of the other will be 1.6x.
Now, $x + 1.6x = 26$
or, $2.6x = 26$
$\therefore x = \frac{260}{26} =$ ₹10
$\therefore$ Price of the other book $= 1.6 \times 10 =$ ₹16

74. 2; Let there be x men.
After 15 days, 500 men are added.
Now, remaining days $= 60 - 15 = 45$
Then, $45 \times x = (x + 500) \times 40$
or, $5x = 20000$
$\therefore x = 4000$

75. 1;

MP		SP	CP
100	$\xrightarrow{-10\%}$	90	?

CP $= 90\left(\frac{100}{120}\right) =$ ₹75
Again, if 15% commission is given
SP = ₹85
$\therefore$ Gain $= 85 - 75 =$ ₹10
$\therefore$ Percentage gain $= \frac{10}{75} \times 100 = 10 \times \frac{4}{3}$
$= \frac{40}{3}\%$

76. 3; For dividing 12 into two natural numbers the sum of the terms of the ratio must be a factor of 12. So, they can't be in the ratio of 3 : 2 because 3 + 2 = 5, which is not a factor of 12.

77. 5;

	Re 1 : 50P : 25P
Ratio no. of coins	2 : 3 : 4
Ratio of values of coins	$2 \times 1 : 3 \times \frac{1}{2} : 4 \times \frac{1}{4}$
	$= 2 : \frac{3}{2} : 1$
	$= 4 : 3 : 2$

$\therefore$ Value of 50P coins $=$ ₹$\frac{3}{9} \times 216 =$ ₹72
Hence the number of 50P coins $= 72 \times 2 = 144$

78. 1; Ratio of speed $\rightarrow$ B : C = 3 : 1
Ratio of time $\rightarrow$ B : C = 1 : 3
(As speed and time are inversely proportional)
$\therefore$ Time taken by B $= \frac{42}{3} \times 1 = 14$ minutes)

79. 4; Cost of 2 dozen mangoes = ₹32
SP of 18 mangoes = 1.5 dozen at ₹12
Then $1.5 \times 12 =$ ₹18
SP of 6 mangoes = ₹$0.5 \times 4 =$ ₹2
$\therefore$ Total SP of 2 dozen mangoes = ₹(18 + 2) = ₹20
Loss $= 32 - 20 = 12$
$\therefore$ Loss % $= \frac{12}{32} \times 100 =$ ₹37.5%

80. 2; SP of 1 kg mixture = ₹9.24; gain 10%
$\therefore$ CP $= \frac{100}{110} \times 9.24 =$ ₹8.4

By alligation method:

CP of 1 kg sugar of 1st kind $\quad$ CP of 1 kg sugar of 2nd kind
9 $\quad$ 7
8.40
1.4 $\quad$ 0.6
7 : 3

Now, 7 : 3 = x : 27
$\therefore x = \frac{27 \times 7}{3} = 63$ kg

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